

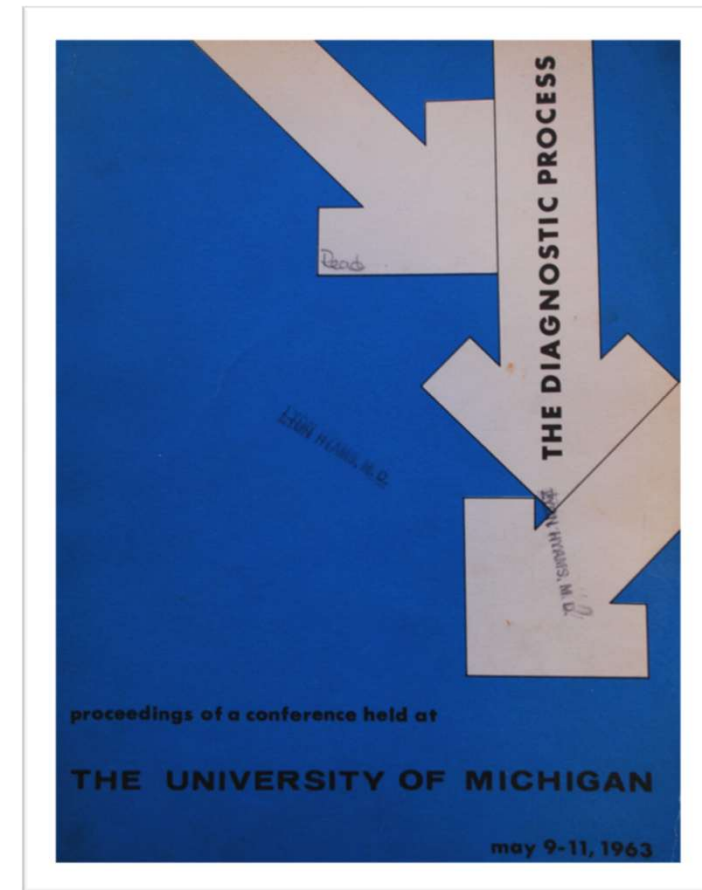
# Mark Iscoe, MD, MHS

UVA Informatics Elective

January 29, 2026

# About me

- From Washington, DC
- BA, Yale College (History)
- Writer at American Museum of Natural History
- MD, Johns Hopkins
- Emergency Medicine residency, NYU/Bellevue
- MHS/CI fellowship, Yale
- Assistant Professor of Biomedical Informatics and Data Science & Emergency Medicine



# Areas of research

- Big picture:
  - Diagnostic process
  - Clinical reasoning
  - Care for older adults
  - Substance use disorders
- Methods:
  - EHR audit log analysis
  - Real-world data
  - Predictive modeling
  - Natural language processing
  - Clinical decision support



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**Research Letter** | Emergency Medicine

## Benchmarking Emergency Physician EHR Time per Encounter Based on Patient and Clinical Factors

Mark S. Iscoe, MD, MHS; Arjun K. Venkatesh, MD, MBA, MHS; Margaret L. Holland, PhD, MPH; Harlan M. Krumholz, MD, SM;  
Karen Dorsey Sheares, MD, PhD; Edward R. Melnick, MD, MHS

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### Introduction

Electronic health records (EHRs) occupy a substantial proportion of emergency physicians' clinical time.<sup>1</sup> In ambulatory care, increased EHR time has been associated with improved care quality<sup>2</sup> but also increased burnout.<sup>3</sup> As emergency department (ED) crowding and acuity increase,<sup>4</sup> expectations for care efficiency and complexity are anticipated to increase. We hypothesized that ED physicians' EHR time varies based on patient and clinical factors, motivating us to benchmark encounter-level factors associated with ED physicians' EHR use.

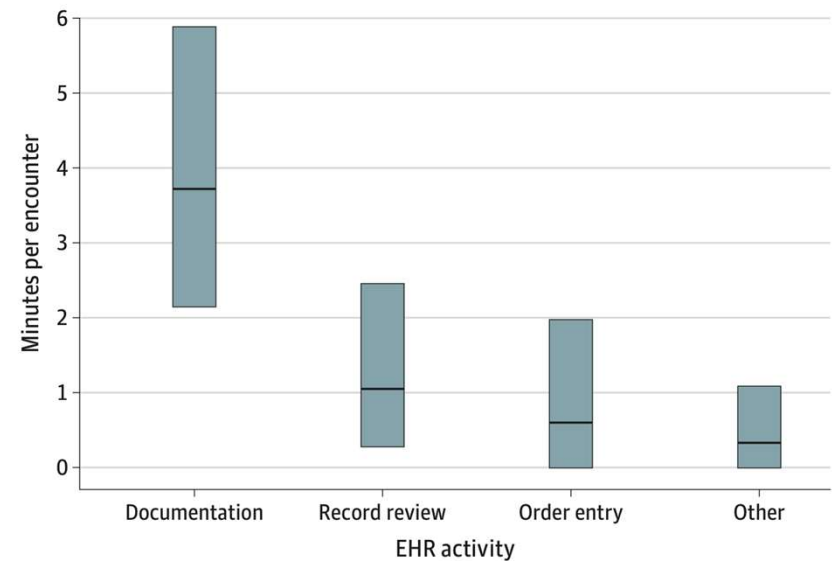
[+ Invited Commentary](#)

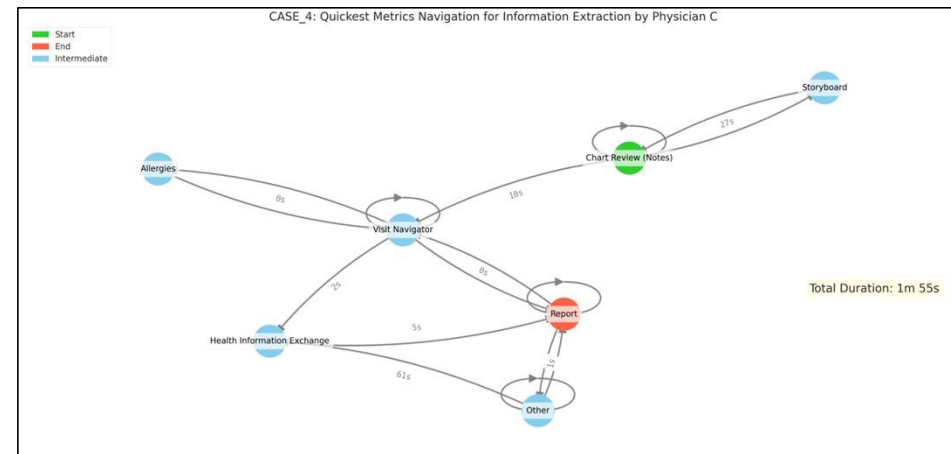
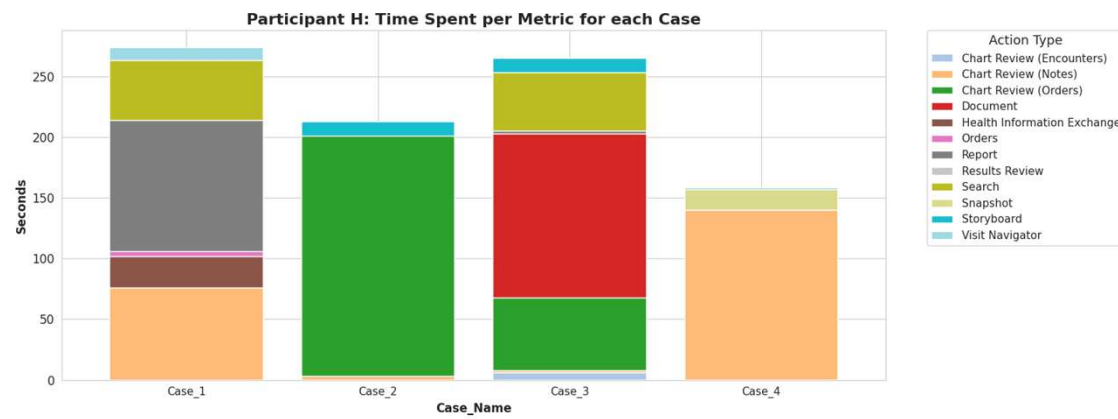
[+ Supplemental content](#)

Author affiliations and article information are listed at the end of this article.

**Table. Patient and Clinical Characteristics and Linear Regression**

| Factor                                                          | Encounters, No. (%)<br>(N = 35 187) | Linear regression<br>$\beta$ (95% CI), min | P value |
|-----------------------------------------------------------------|-------------------------------------|--------------------------------------------|---------|
| Patient                                                         |                                     |                                            |         |
| Age, Mean (SD) y                                                | 54.7 (20.6)                         | 0.005 (0.002-0.008)                        | .001    |
| Sex                                                             |                                     |                                            |         |
| Female                                                          | 18 953 (53.9)                       | 0 [Reference]                              | .30     |
| Male                                                            | 16 234 (46.1)                       | -0.04 (-0.12 to 0.04)                      |         |
| Self-reported race and ethnicity from EHR database <sup>a</sup> |                                     |                                            |         |
| American Indian or Alaska Native                                | 58 (0.2)                            | -0.28 (-0.95 to 0.39)                      | .35     |
| Asian                                                           | 660 (1.9)                           | -0.07 (-0.31 to 0.17)                      |         |
| Black or African American                                       | 7886 (22.4)                         | -0.08 (-0.17 to 0.02)                      |         |
| Hispanic, Latina, Latino, or Latinx                             | 5934 (16.9)                         | -0.07 (-0.20 to 0.05)                      |         |
| Multiracial                                                     | 205 (0.6)                           | -0.05 (-0.40 to 0.30)                      |         |
| Native Hawaiian or Other Pacific Islander                       | 36 (0.1)                            | -0.49 (-1.51 to 0.53)                      |         |
| White                                                           | 19 538 (55.5)                       | 0 [Reference]                              |         |
| Other or not listed <sup>b</sup>                                | 870 (2.5)                           | -0.07 (-0.30 to 0.16)                      |         |
| Emergency Severity Index <sup>c</sup>                           |                                     |                                            |         |
| 1                                                               | 315 (0.9)                           | 2.83 (1.89 to 3.76)                        | <.001   |
| 2                                                               | 12 418 (35.3)                       | 1.37 (0.88 to 1.85)                        |         |
| 3                                                               | 18 850 (53.6)                       | 1.10 (0.64 to 1.57)                        |         |
| 4                                                               | 3166 (9.0)                          | 0.33 (-0.21 to 0.68)                       |         |
| 5                                                               | 438 (1.2)                           | 0 [Reference]                              |         |
| Disposition                                                     |                                     |                                            |         |
| Admission                                                       | 9967 (28.3)                         | 0.52 (0.37 to 0.67)                        | <.001   |
| Departure against medical advice                                | 358 (1.0)                           | 1.85 (0.92 to 2.77)                        |         |
| Discharge                                                       | 22 023 (62.6)                       | 0 [Reference]                              |         |
| ED observation                                                  | 2371 (6.7)                          | 0.49 (0.23 to 0.76)                        |         |
| Other <sup>b</sup>                                              | 468 (1.3)                           | 0.60 (0.17 to 1.03)                        |         |
| Chief concern                                                   |                                     |                                            |         |
| Abdominal pain                                                  | 3552 (10.1)                         | 0.21 (-0.11 to 0.53)                       | <.001   |
| Alcohol intoxication                                            | 592 (1.7)                           | 0 [Reference]                              |         |
| Back pain                                                       | 647 (1.8)                           | 0.83 (0.44 to 1.23)                        |         |
| Chest pain                                                      | 2618 (7.4)                          | 0.42 (0.10 to 0.74)                        |         |
| Dizziness                                                       | 886 (2.5)                           | 0.76 (0.42 to 1.09)                        |         |
| Fall                                                            | 1384 (3.9)                          | 0.79 (0.44 to 1.14)                        |         |
| Motor vehicle crash                                             | 800 (2.3)                           | 0.83 (0.43 to 1.23)                        |         |
| Shortness of breath                                             | 1859 (5.3)                          | 0.69 (0.39 to 1.00)                        |         |
| Other <sup>b</sup>                                              | 22 849 (64.9)                       | 0.43 (0.13 to 0.73)                        |         |
| Clinical                                                        |                                     |                                            |         |
| Resident or APP assigned                                        | 19 623 (55.8)                       | 0.13 (0.07 to 0.33)                        | .21     |
| Handoff between ED attending physicians                         | 13 656 (38.8)                       | 2.05 (1.92 to 2.17)                        | <.001   |
| Practice setting                                                |                                     |                                            |         |
| Academic ED                                                     | 16 364 (46.5)                       | 0 [Reference]                              | <.001   |
| Community ED                                                    | 12 076 (34.3)                       | 1.31 (0.71 to 1.91)                        |         |
| Freestanding ED                                                 | 6747 (19.2)                         | 1.08 (0.62 to 1.54)                        |         |





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RESEARCH METHODOLOGY: EMPIRICAL  
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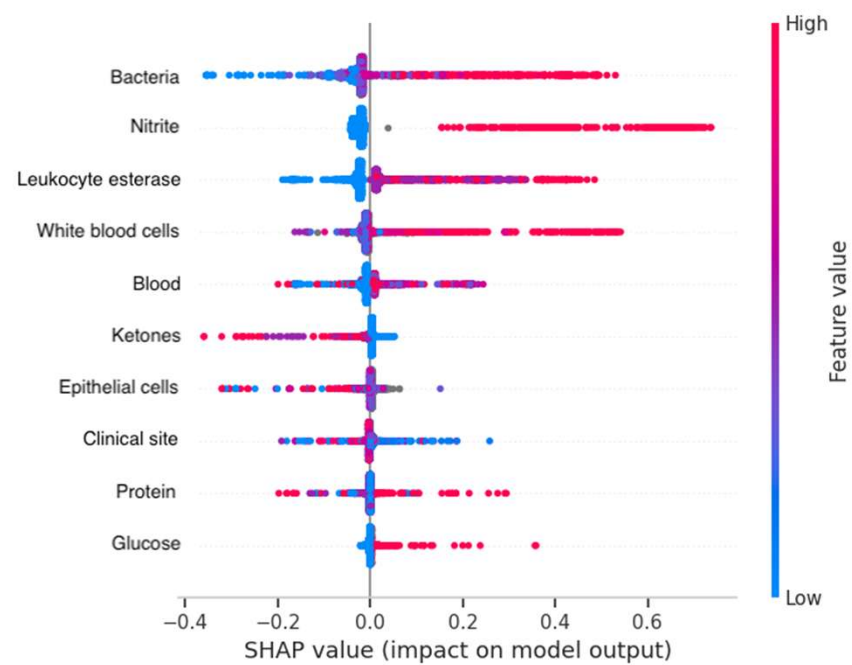
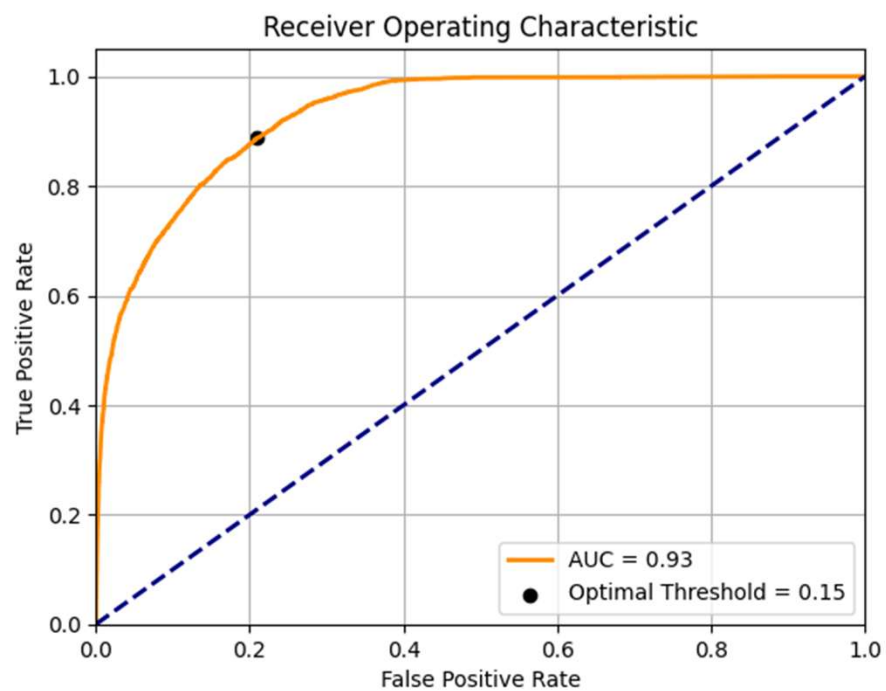
# Identifying signs and symptoms of urinary tract infection from emergency department clinical notes using large language models

Mark Iscoe MD, MHS<sup>1,2</sup> | Vimig Socrates MS<sup>2,3</sup> | Aidan Gilson BS<sup>4</sup> | Ling Chi BS<sup>5</sup> |  
Huan Li MS<sup>3</sup> | Thomas Huang BS<sup>4</sup> | Thomas Kearns MD<sup>1</sup> | Rachelle Perkins MD<sup>1</sup> |  
Laura Khandjian MD<sup>1</sup> | R. Andrew Taylor MD, MHS<sup>1,2</sup>

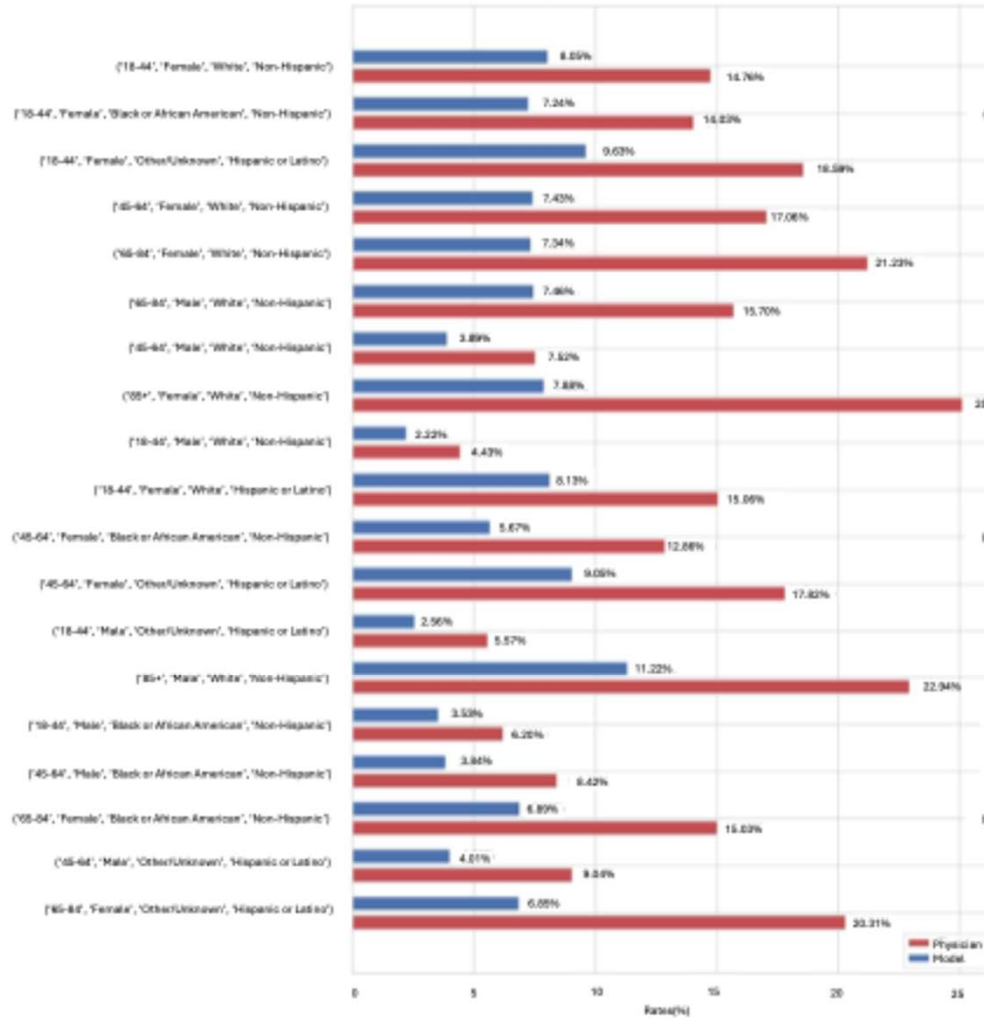
**Legend:** ■ B-Dysuria (burning) ■ B-Urinary\_frequency (urinary) ■ B-Dysuria (discomfort)

## Word Importance

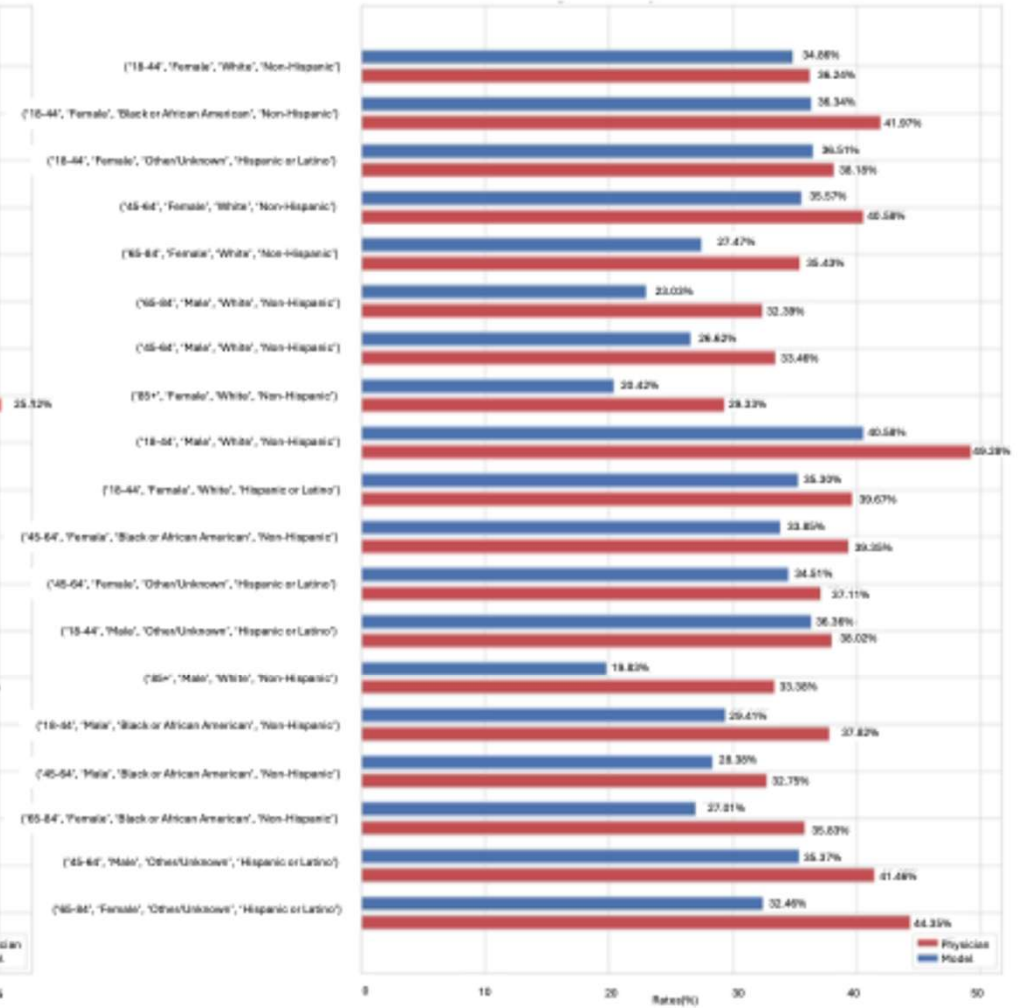
#s The patient, a [REDACTED], presents to the ED primarily complaining of dysuria, which has persisted for the past 3 days. The dysuria is described as a sharp, burning sensation during urination, progressively intensifying since its onset. Additionally, the patient reports experiencing increased urinary frequency, with an urge to void approximately every 1 - 2 hours, and this has notably disrupted nocturnal sleep. The patient also notes the urine to be cloudy in appearance and accompanied by an offensive odor. Suprapubic pain, characterized as a constant, dull ache rated 4 / 10 on the pain scale, was noted to begin approximately 2 days ago. The pain is localized, non-radiating, and not evidently linked to particular activities or dietary patterns. Significantly, the patient explicitly denies experiencing nausea, vomiting, or any alterations in mental status. There have been no episodes of fever, chills, or flank pain. No recent sexual activity, changes in sexual partners, or history of sexually transmitted infections were reported. The patient has not used antibiotics recently and has no known history of urinary tract infections or kidney stones. No recent travel or exposure to individuals with similar symptoms was reported. In terms of management, the patient attempted to alleviate symptoms by increasing fluid intake and using over-the-counter pain relief (acetaminophen) with only marginal relief. The decision to seek care in the ED was prompted by the persistence and gradual worsening of symptoms, particularly the discomfort during urination and the ongoing suprapubic pain. #/s



Overdiagnosis Rates – Physician Behavior vs Model Performance



Underdiagnosis Rates – Physician Behavior vs Model Performance



## Case Report

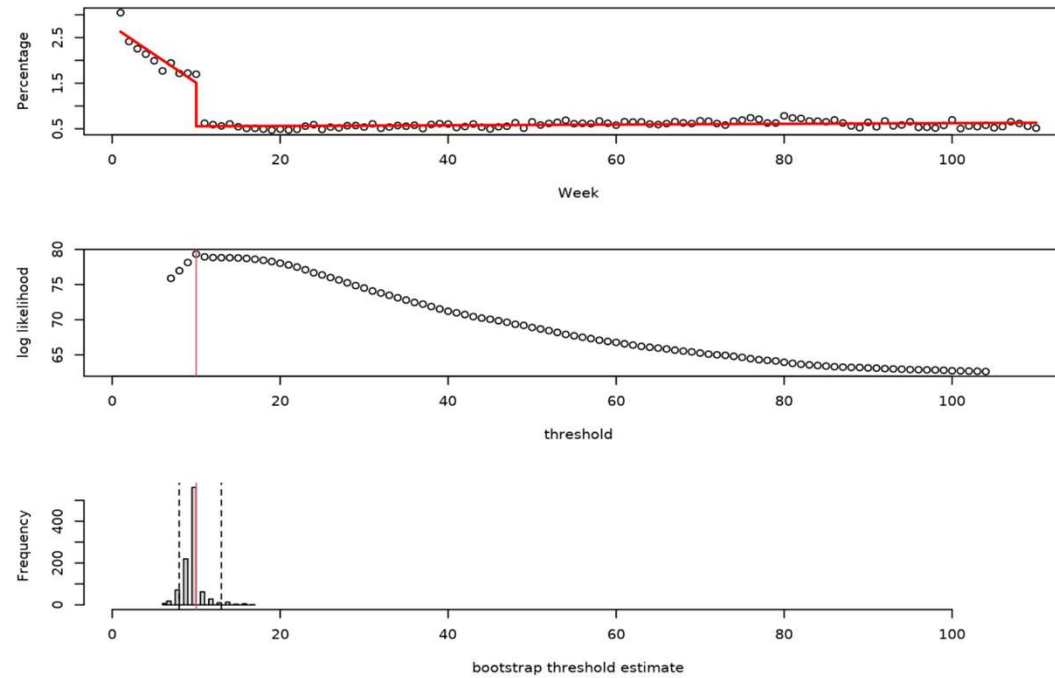
# Case Report: A health system's experience using clinical decision support to promote note sharing after the 21st Century Cures Act

Mark Iscoe, MD, MHS<sup>\*,1,2</sup>, Arjun K. Venkatesh , MD, MBA, MHS<sup>2,3</sup>, Emily M. Powers, MD, MHS<sup>4</sup>, Nitu Kashyap, MD<sup>5</sup>, Allen L. Hsiao, MD<sup>1,2,4</sup>, Hun Millard, MD<sup>6,7</sup>, Rohit B. Sangal , MD, MBA<sup>2</sup>

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**Figure 2.** Step-segmented threshold regression model predicting rates of unshared notes by intervention week. The top plot shows a scatterplot of unshared note rates by intervention week, with the identified threshold after week 10, when the hard-stop alert requiring a reason for not sharing notes was introduced. The red lines indicate the linear regression trends before and after this threshold and the stepwise drop at the threshold. The middle and bottom plot show threshold identification by likelihood ratio (log) and bootstrap threshold estimate distribution frequency, respectively; both identify week 10 as the most likely threshold.

# A Measurement Science Framework to Optimize CDS for Opioid Use Disorder Treatment in the ED

Mark S. Iscoe<sup>1,2</sup> Carolina Diniz Hooper<sup>1</sup> Deborah R. Levy<sup>2,3</sup> John Lutz<sup>4</sup> Hyung Paek<sup>5</sup>  
Christian Rose<sup>6</sup> Thomas Kannampallil<sup>7</sup> Daniella Meeker<sup>2</sup> James D. Dziura<sup>1,5</sup> Edward R. Melnick<sup>1,2,5</sup>

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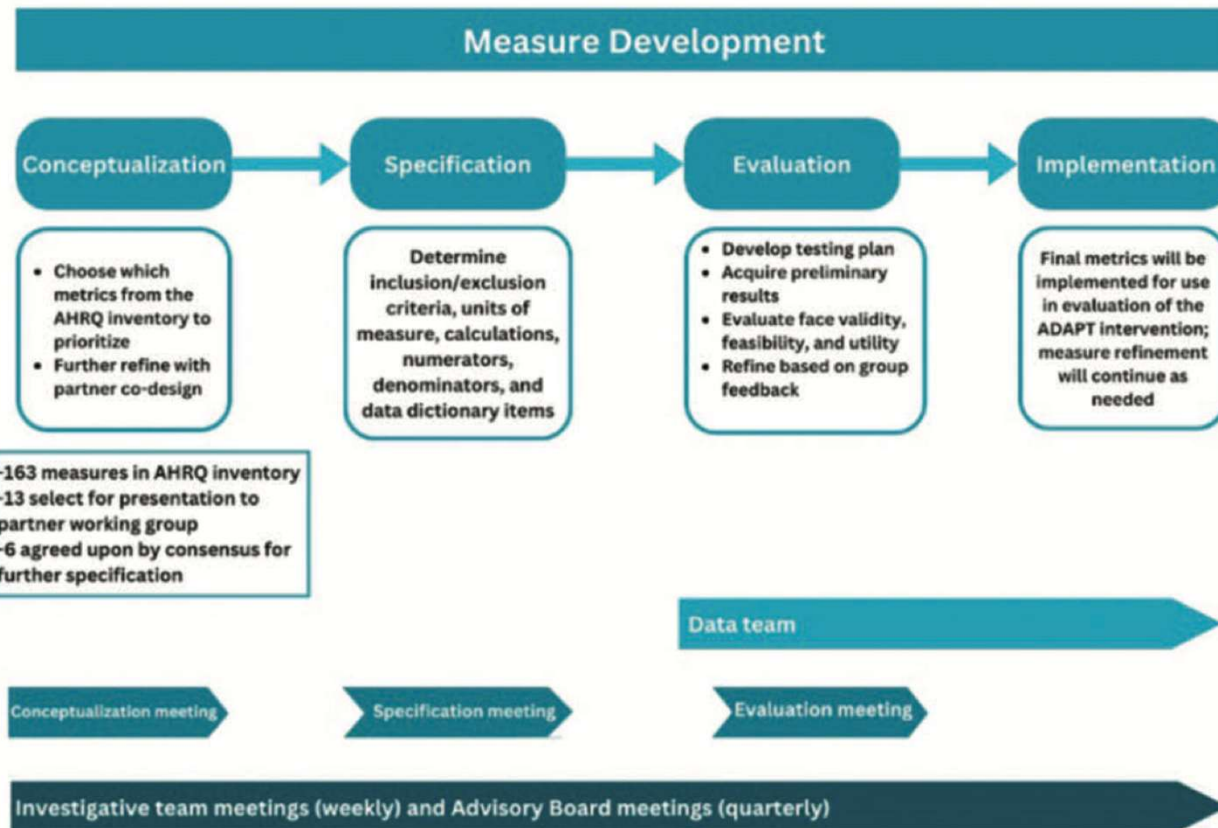
<sup>5</sup>Department of Biostatistics (Health Informatics), Yale School of Public Health, New Haven, Connecticut, United States

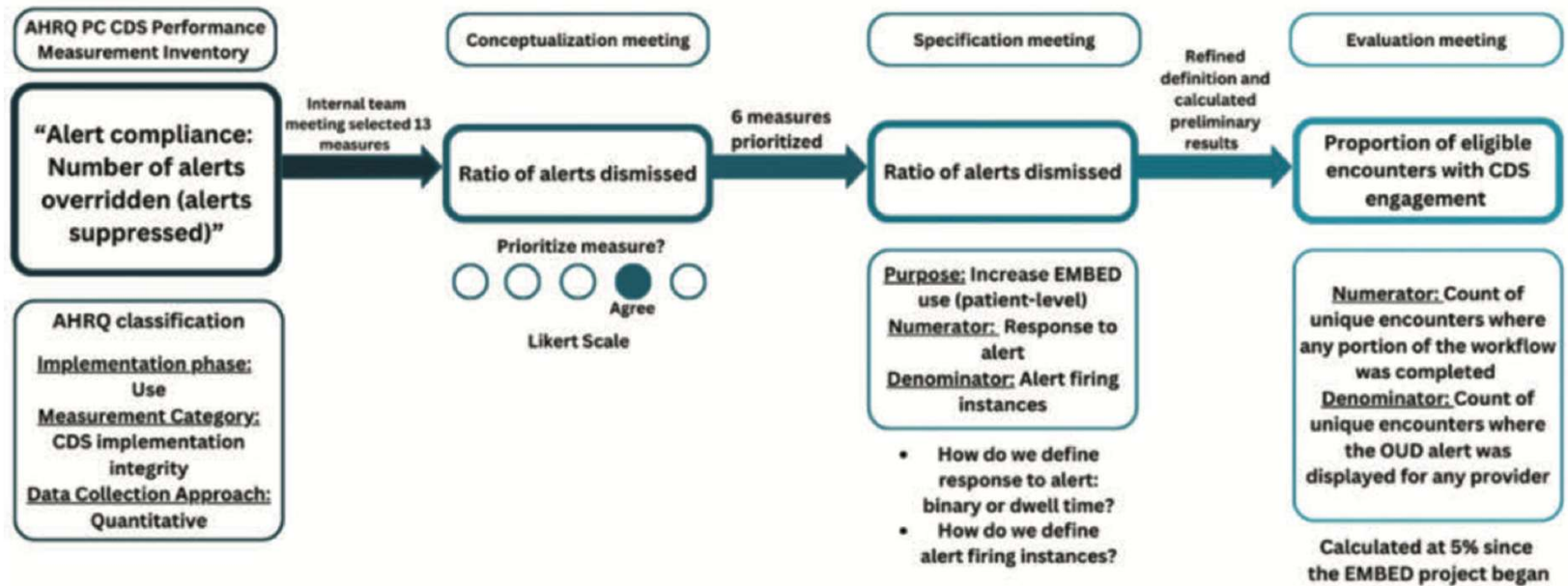
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**Table 2** Final measures along with intended purpose, the abbreviated formula for calculation, and preliminary value based on the use of the current EMBED CDS from January 1, 2023, to December 31, 2024

| Measure                                               | EMBED shortcoming addressed                                                                               | Purpose                                 | Formula (brief)                                                                                                                                                                                                                                                                                | Baseline value (95% CI) |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Proportion of eligible encounters with CDS engagement | Infrequent CDS use (launched in only 9.4% of eligible encounters in the initial EMBED trial) <sup>a</sup> | Increase CDS engagement                 | Numerator: Encounters in which any portion of the EMBED workflow (OUD diagnosis, withdrawal assessment, readiness assessment, buprenorphine or adjunct medication order/prescription, postvisit referral order) was documented or completed<br>Denominator: Encounters in which an alert fired | 5.0% (4.3–5.8%)         |
| Teamwork on ED initiation of buprenorphine            | No direct involvement of nursing in CDS workflow <sup>b</sup>                                             | Increase collaborative, team-based care | Numerator: Encounters in which more than one team member (regardless of role) helped complete the EMBED workflow (by definition above)<br>Denominator: Encounters in which at least one user engaged with the EMBED CDS (see comment above)                                                    | 39.9% (32.5–47.3%)      |
| Proportion of eligible users who used EMBED           | Poor reach (launched by only 38.2% of physicians in the initial EMBED trial) <sup>a</sup>                 | Increase EMBED use on a user level      | Numerator: Staff members across roles who used EMBED (by definition above) at least once during the study period<br>Denominator: Staff members who cared for at least one patient for whom the alert fired during the study period                                                             | 58.3% (50.9–65.8%)      |



Cosmos is a dataset created in collaboration with a community of health systems using Epic and is designed to improve patient care. By combining their data, participating organizations and Epic can make new discoveries and advance medicine. Cosmos also powers tools at the point of care, providing insights to clinicians that are tailored to the patient in front of them.

[More about Cosmos →](#)

# Cosmos: PIM prescriptions in older adults

| Characteristic                                    | Overall          | 65-74          | 75-84          | 85-94          | 95+          |
|---------------------------------------------------|------------------|----------------|----------------|----------------|--------------|
|                                                   | N = 16,130,160*  | N = 7,808,102* | N = 5,593,169* | N = 2,409,984* | N = 318,905* |
| <b>PIM Prescriptions</b>                          |                  |                |                |                |              |
| Any PIM                                           | 1,055,673 (6.5%) | 648,502 (8.3%) | 319,436 (5.7%) | 82,152 (3.4%)  | 5,583 (1.8%) |
| Barbiturate                                       | 22,864 (0.1%)    | 14,923 (0.2%)  | 6,422 (0.1%)   | 1,434 (<0.1%)  | 85 (<0.1%)   |
| Benzodiazepine                                    | 50,818 (0.3%)    | 30,039 (0.4%)  | 16,272 (0.3%)  | 4,184 (0.2%)   | 323 (0.1%)   |
| First-Generation Antihistamine                    | 336,947 (2.1%)   | 192,913 (2.5%) | 110,492 (2.0%) | 31,396 (1.3%)  | 2,146 (0.7%) |
| First-Generation Antipsychotic                    | 19,475 (0.1%)    | 12,218 (0.2%)  | 5,586 (<0.1%)  | 1,493 (<0.1%)  | 178 (<0.1%)  |
| Metoclopramide                                    | 37,553 (0.2%)    | 22,940 (0.3%)  | 11,346 (0.2%)  | 3,052 (0.1%)   | 215 (<0.1%)  |
| Nonbenzodiazepine Benzodiazepine Receptor Agonist | 2,808 (<0.1%)    | 1,613 (<0.1%)  | 911 (<0.1%)    | 264 (<0.1%)    | 20 (<0.1%)   |
| Skeletal Muscle Relaxant                          | 599,046 (3.7%)   | 383,112 (4.9%) | 172,284 (3.1%) | 41,015 (1.7%)  | 2,635 (0.8%) |
| Sulfonylurea                                      | 5,326 (<0.1%)    | 3,438 (<0.1%)  | 1,440 (<0.1%)  | 414 (<0.1%)    | 34 (<0.1%)   |
| * Mean (SD); n (%)                                |                  |                |                |                |              |

# Cosmos: PIM prescriptions in older adults

**Table 2. Distribution and variation of emergency departments' PIM prescribing rates, overall and by medication class, in descending order of prescription frequency.**

| Medication class                                  | Median department-level prescription rate (IQR) | Coefficient of variation* (CV) across departments |
|---------------------------------------------------|-------------------------------------------------|---------------------------------------------------|
| Any PIM                                           | 5.89% (4.44-7.86)                               | 0.48                                              |
| Skeletal Muscle Relaxant                          | 3.14% (2.07-4.53)                               | 0.64                                              |
| First-Generation Antihistamine                    | 1.92% (1.44-2.46)                               | 0.61                                              |
| Benzodiazepine                                    | 0.25% (0.14-0.41)                               | 1.03                                              |
| Metoclopramide                                    | 0.18% (0.10-0.29)                               | 1.05                                              |
| First-Generation Antipsychotic                    | 0.09% (0.04-0.16)                               | 2.01                                              |
| Barbiturate                                       | 0.05% (0-0.15)                                  | 1.85                                              |
| Sulfonylurea                                      | 0.02% (0-0.04)                                  | 2.88                                              |
| Nonbenzodiazepine Benzodiazepine Receptor Agonist | 0.01% (0-0.02)                                  | 5.41                                              |

\* The coefficient of variation is the ratio of the standard deviation to the mean; higher values suggest greater variation.

# Generative Medical Event Models Improve with Scale

Shane Waxler<sup>\*1</sup>, Paul Blazek<sup>\*1</sup>, Davis White<sup>\*1</sup>, Daniel Sneider<sup>\*1</sup>,  
Kevin Chung<sup>1</sup>, Mani Nagarathnam<sup>1</sup>, Patrick Williams<sup>1</sup>, Hank Voeller<sup>1</sup>, Karen Wong<sup>1</sup>,  
Matthew Swanhorst<sup>1</sup>, Sheng Zhang<sup>2</sup>, Naoto Usuyama<sup>2</sup>, Cliff Wong<sup>2</sup>, Tristan Naumann<sup>2</sup>,  
Hoifung Poon<sup>2</sup>, Andrew Loza<sup>3</sup>, Daniella Meeker<sup>3, 4</sup>, Seth Hain<sup>1</sup>, and Rahul Shah<sup>†1</sup>

<sup>1</sup>Epic Systems

<sup>2</sup>Microsoft Research

<sup>3</sup>Yale School of Medicine

<sup>4</sup>Cosmos Governing Council

## Abstract

Realizing personalized medicine at scale calls for methods that distill insights from longitudinal patient journeys, which can be viewed as a sequence of medical events. Foundation models pretrained on large-scale medical event data represent a promising direction for scaling real-world evidence generation and generalizing to diverse downstream tasks. Using Epic Cosmos, a dataset with medical events from de-identified longitudinal health records for 16.3 billion encounters over 300 million unique patient records from 310 health systems, we introduce the Curiosity models, a family of decoder-only transformer models pretrained on 118 million patients representing 115 billion discrete medical events (151 billion tokens). We present the largest scaling-law study of medical event data, establishing a methodology for pretraining and revealing power-law scaling relationships for compute, tokens, and model size. Consequently, we pretrained a series of compute-optimal models with up to 1 billion parameters. Conditioned on a patient’s real-world history, Curiosity autoregressively predicts the next medical event to simulate patient health timelines. We studied 78 real-world tasks, including diagnosis prediction, disease prognosis, and healthcare operations. Remarkably for a foundation model with generic pretraining and simulation-based inference, Curiosity generally outperformed or matched task-specific supervised models on these tasks, without requiring task-specific fine-tuning or few-shot examples. Curiosity’s predictive power consistently improves as the model and pretraining scale. Our results show that Curiosity, a generative medical event foundation model, can effectively capture complex clinical dynamics, providing an extensible and generalizable framework to support clinical decision-making, streamline healthcare operations, and improve patient outcomes.

# Ongoing projects

- Real-time predictive modeling to inform CDS:
  - Urine culture prediction (YNHH Innovation; Huan Li)
  - Dementia diagnosis (NIA; Inna Cohen, Emily Johns)
- NLP:
  - Identification of adverse drug reactions in unstructured clinical notes (NCATS KL2; Yuan Gao)
- EHR use patterns:
  - Exploration of ED physicians' information-gathering in simulated cases (AMA; Varada Khanna)
- Epic Cosmos:
  - Various observational studies related to older adults, medication safety, ED care, and opioid use disorder (Yu Sun, Rie Sakai)
- GEAR (Geriatric Emergency Care Applied Research) Network:
  - Boarding in the ED
- Ambient AI and clinical notes:
  - Analysis of note quality and completeness with and without AI (Chartash)
- Collaborators/mentors: Ted Melnick, Daniella Meeker, Andrew Taylor (UVA), Lucila Ohno-Machado, Hua Xu, Qingyu Chen

# Lab meetings

- Tuesdays, 3-4pm
- Virtual (Teams) or in-person
- Email me ([mark.iscoe@yale.edu](mailto:mark.iscoe@yale.edu)) if you'd like to join